RASHAD HAMIDI

PhD Researcher in Mathematical and Theoretical Physics - Durham University
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RESEARCH PROFILE & INTERESTS

Core Research Areas: Integrable Field Theory | String Theory | Quantum Field Theory

My primary research interests lie in the areas of integrable systems, string theory and quantum field theory. I am particularly focused on exploring integrable sigma models, their deformations, and their connections to higher-dimensional gauge theories, such as 4d Chern-Simons theory. My work involves studying the mathematical structures underlying these models, including quantum groups and Yang-Baxter equations, as well as investigating their physical applications.

Vision: My long-term research vision is to implement new technologies such as artificial intelligence, machine learning and quantum computing in theoretical physics research, and use them to discover new integrable field theories.

EDUCATION & ACADEMIC BACKGROUND

PhD in Mathematical and Theoretical Physics *Nov 2022 - Aug 2026*
Durham University, Department of Mathematical Sciences - Durham, UK
Supervisor: Prof. Ben Hoare
Thesis: "Integrable Sigma Models with Twisted Quantum Affine Symmetry"
Focus: Integrable Field Theory, String Theory, Classical Integrability, Sigma Models, R-Matrix, Quantum Affine Symmetry, Yang-Baxter Equation, Integrable Deformations

MSc in Particles, Strings and Cosmology *Oct 2021 - Sep 2022*
Durham University, Department of Mathematical Sciences - Durham, UK
Degree Achievement: Distinction
Supervisor: Prof. Nabil Iqbal
Thesis: "BTZ Black Hole: Holographic Duality, Entropy, and AdS3/CFT2"

BSc in Physics *Sep 2017 - Jun 2021*
Birzeit University, Department of Physics - Birzeit, Palestine
Degree Achievement: Distinction
Supervisor: Prof. Abdallah Sayyed-Ahmad
Thesis: "Fractional Quantum Mechanics"

PUBLICATIONS & PREPRINTS

[1] Rashad Hamidi, Ben Hoare
"Twists of trigonometric sigma models"
JHEP 08 (2025) 090 | arXiv: 2504.18492 [hep-th] | DOI: 10.1007/JHEP08(2025)090

[2] Rashad Hamidi, Ben Hoare, Ana Retore
"Twisted quantum affine symmetry and the Z2-twisted eta-model"
(soon) | arXiv: [soon] | DOI:

[3] Rashad Hamidi
"Coupled twisted sigma models"
(soon) | arXiv: [soon] | DOI:

ACADEMIC APPOINTMENTS & WORK EXPERIENCE

AI Physics and Mathematics Trainer *Jun 2026 - Present*
Mercor - Online (project-based)
- Author advanced physics and mathematics problems with step-by-step solutions for model fine-tuning (SFT).
- Evaluate and rank model reasoning outputs, isolating mathematical fallacies and hallucinations.
- Design adversarial edge cases and evaluation rubrics to benchmark complex multi-step reasoning.

Teaching Assistant *Jan 2023 - May 2026*
Durham University - Durham, UK (part time)
Modules taught: Discrete Mathematics, Maths for Engineers and Scientists, Analysis I, Single Math A, Single Math B, Mathematical Physics II, Special Relativity & Electromagnetism II.

Astronomical Observatory Tour Guide *Jan 2018 - Jun 2021*
Birzeit University & Michel and Sanieh Hakim Observatory - Birzeit, Palestine (part time)
- Led engaging and educational guided tours of the observatory facilities for diverse audiences.
- Developed and delivered accessible presentations on astronomical concepts and celestial objects.
- Operated the main telescope to showcase planets, stars, nebulae, and galaxies during public nights.
- Assisted in planning, promotion, and execution of public outreach events (eclipse viewings, meteor showers).

HONOURS, AWARDS & GRANTS

- **UKRI STFC PhD Studentship** *2022 - 2026*
  Durham University | UKRI / STFC - Full doctoral research studentship funded by the Science and Technology Facilities Council (STFC) and UKRI. (Sponsors: UK Research and Innovation (UKRI), Science and Technology Facilities Council (STFC), Durham University)
- **MSc Degree with Distinction** *2022*
  Durham University - Awarded Distinction in MSc in Particles, Strings and Cosmology.

- **Global Masters Scholarship** ***2021 - 2022***
  Durham University - Competitive international merit scholarship covering tuition and stipend. (Sponsors: Durham University, Scott Logic, Trevelyan College)
- **BSc Degree with Distinction** ***2021***
  Birzeit University - Graduated with Distinction in Physics (Major) and Mathematics (Minor).
- **Mosa Naser Scholarship for Physics (3)** ***2018 - 2021***
  Birzeit University | Dept. of Physics - Awarded 3 consecutive academic years for top ranking academic performance in the physics department. (Sponsors: Birzeit University, Department of Physics)
- **Science Faculty Scholarship** ***2018 - 2019***
  Birzeit University | Faculty of Science - Academic merit scholarship awarded by the Faculty of Science. (Sponsors: Birzeit University, Faculty of Science)
- **Paltel Group Scholarship** ***2017 - 2021***
  Paltel Group | Birzeit University - Four-year university scholarship awarded for outstanding pre-university academic excellence. (Sponsors: Birzeit University, Paltel Group)
- **University Honor Scholarship (7 Semesters)** ***2018 - 2021***
  Birzeit University - Awarded for 7 semesters for holding top cumulative and semester GPA. (Sponsors: Birzeit University)
- **Dean's Honor Roll (8 Consecutive Semesters)** ***2017 - 2021***
  Birzeit University - Maintained Honor Roll distinction every single semester of undergraduate studies.

RESEARCH GROUPS & AFFILIATIONS

- **Ben Hoare Integrability Group**, *Durham University* ***2022 - 2026***
- **Mathematical and Theoretical Physics Group**, *Durham University* ***2022 - 2026***
- **High Energy Physics for Palestine Group (HEP4P)**, *(remote collaboration)* ***2022 - Present***
- **Centre for Particle Theory**, *Durham University* ***2021 - 2026***
- **Physics Club**, *Birzeit University* ***2017 - 2021***
- **Astronomy Club**, *Birzeit University* ***2017 - 2021***

CONFERENCES & SYMPOSIA

- **Integrability, Dualities and Deformations Conference:** Wroclaw (2026), Stockholm [Poster] (2025), Swansea (2024), Durham (2023)
- **Integrability in Gauge and String Theory:** DESY Hamburg (2026), King's College London (2025), ETH Zurich (2023)
- **Integrability in Quantum Field Theory:** Durham (2026)

SEMINARS & COLLOQUIA

- **North British Mathematical Physics Seminars (NBMPS):** 75th (Durham, 2025), 74th [Speaker] (York, 2025), 72nd (Durham, 2024), 71st (York, 2024), 69th (Durham, 2023), 66th (Durham, 2022)
- **Physics Department Seminars:** Meteorology (Birzeit, 2020), Computational Multiphysics in Condensed Matter (Birzeit, 2019), Black Holes (Birzeit, 2017)

ADVANCED SCHOOLS & WORKSHOPS

- **Paths to Quantum Field Theory Workshop:** Durham (2023), Durham (2022)
- **Young Researchers Integrability School and Workshop:** Durham (2023)
- **Hands-on Molecular Dynamics Workshop:** Birzeit (2019)
- **Palestinian Advanced Physics School:** Birzeit (2019)
- **Physics Without Frontiers Workshop:** Birzeit (2017)

COMPETITIONS & CERTIFICATION PROGRAMS

- **Durham Hackathon (DurHack)** - *Durham University* ***Nov 2024***
- **Pre-University Competitions:** Science Exhibition (2015), Physics Olympiad (2015), First Lego League (FLL) Robotics (2015), First Lego League (FLL) Robotics (2014), Intel ISEF (2012), SyscoLab Competition (2012), SEEK Competition (2012) (Palestine, 2012-2015).
- **AMIDEAST Access Program** - *Ramallah* ***2014-2016***
- **Science and Technology Entrepreneurship Program (STEP)**- *Ramallah* ***2014-2015***

TECHNICAL & SCIENTIFIC SKILLS

- **Programming (main):** Python, Mathematica
- **Programming (familiar):** C, R, Matlab, Octave
- **Scientific Writing:** LaTeX, Jupyter Notebook, Mathematica Notebook, Markdown
- **Scientific Computing Skills:** Numerical Methods, Statistical Analysis
- **Electronics:** Arduino, Raspberry Pi, LabVIEW, NXT Robotics, SEEK Electronic Bag, Syscolab Electronic Bag, Sophene Electronic Bag
- **Astronomy:** Celestia, Stellarium, RSpec
- **Web Design:** HTML, CSS, JavaScript, TypeScript
- **Modeling:** Unity, Blender, Unreal Engine, AutoCAD
- **Languages:** Arabic: Native, English: Fluent

DEVELOPED APPLICATIONS & SCIENTIFIC DATASETS

Apps: https://xurashad.github.io/apps | ***Datasets:*** https://xurashad.github.io/datasets

- **Scientific Applications:** Particle Physics Visualizer, Interactive Periodic Table, Fractal Explorer, Feynman Diagram Generator, Graphing Calculator, Celestial Simulator, Earth Explorer
- **Datasets Generators:** Fibonacci Sequence Generator, Prime Sequence Generator

- **Professional Applications:** Website Builder Pro, CV Builder Pro, LaTeX Editor Pro, Office Suite Pro
- **AI Applications:** AI Chatbot, ArtForge AI Image Generator
- **Utilities:** Plagiarism Checker, Unicode Explorer, Unicode Font Generator, PDF Tools, Multimedia Base64 Converter, QR Code Generator, Image Editor
- **Games:** 2048 Tile, Celestial Chess
- **Curated Datasets:** Fundamental & Supersymmetric Particles, First 1M Prime Numbers, Chemical Elements

COURSE-BASED & TECHNICAL PROJECTS

- **"BTZ Black Hole: Holographic Duality, Entropy, and AdS3/CFT2"** *Particles, Strings and Cosmology (MSc Dissertation) (Summer 2022)*
- **"Fractional Quantum Mechanics"** *Quantum Mechanics (Spring 2021)*
- **"Computational Planetary Motion"** *Computational Physics (Fall 2020)*
- **"Many Body Problem"** *Computational Physics (Fall 2020)*
- **"Cellular Automata"** *Computational Physics (Fall 2020)*
- **"Interactive Visualization of the Logistic Map"** *Computational Physics (Fall 2020)*
- **"Stellar Spectroscopy of Rigil, Betelgeuse and Sirius"** *Astronomy (Spring 2020)*
- **"Magnetic Fluids"** *Nanophysics (Fall 2019)*
- **"Vantablack: Optical Coating and 3D Light Microscopy"** *Medical Physics (Spring 2019)*

COMPREHENSIVE ACADEMIC COURSEWORK

Durham University - MSc in Particles, Strings and Cosmology:

- **Quantum Field Theory:** Introductory Field Theory, Quantum Field Theory I, Quantum Field Theory II, Quantum Electrodynamics, Supersymmetry
- **General Relativity and Cosmology:** General Relativity, Cosmology I, Cosmology II, Neutrino Physics, Astroparticle Physics
- **Gauge Field Theory:** Group Theory, Standard Model, Renormalisation Group, Scattering Amplitudes, Non-Perturbative Physics
- **Superstring Theory:** Conformal Field Theory, String Theory
- **Phenomenology:** Quantum Chromodynamics, Flavour Physics and Effective Field Theories, Higgs Physics
- **Extra:** Introduction to AdS/CFT, Analytic loop integration

Birzeit University - BSc in Physics (Minor Mathematics):

- **Physics:** General Physics I--III, Modern Physics, Medical Physics, Astronomy, Vibrations and Waves, Optics, Analog Electronics, Mathematical Physics, Classical Mechanics I--III, Electromagnetic Theory I--II, Quantum Mechanics I--II, Thermal Physics & Statistical Mechanics, Computational Physics
- **Physics Labs:** General Physics Lab I--III, Modern Physics Lab, Analogue Electronics Lab, Advanced Physics Lab
- **Mathematics:** Calculus I--III, Foundations of Mathematics (Logic), Linear Algebra I--II, Ordinary Differential Equations, Partial Differential Equations, Numerical Methods, Abstract Algebra I, Mathematical Analysis I
- **Statistics and Computer Science:** Statistics I, Probability Theory, C Programming
- **Extra Scientific Courses:** General Chemistry I, General Biology I
- **Extra Scientific Labs:** General Chemistry Lab I, General Biology Lab I
- **Philosophy:** Modern and Contemporary European Civilisation, Modern and Contemporary Arab Thought

ACADEMIC REFEREES

- **Prof. Ben Hoare** *(PhD supervisor)*
  Durham University | Email: ben.hoare@durham.ac.uk | Web: https://www.benhoare.info/
- **Dr. Ana Retore** *(Groupmate & Collaborator)*
  DESY | Email: ana.retore@desy.de
- **Prof. Nabil Iqbal** *(MSc supervisor)*
  Durham University | Email: nabil.iqbal@durham.ac.uk | Web: https://www.nabiliqbal.com/
- **Prof. Benoit Vicedo** *(Colleague)*
  University of York | Email:b.vicedo@york.ac.uk
- **Prof. Hazem Abusara** *(BSc professor)*
  Birzeit University | Email: habusara@birzeit.edu